Local Area Networks (LANs)

What is a network?

A network can be defined as "two or more computers connected together to share information and resources". This can involve physical or wireless connections, or both.

What is a LAN?

A LAN is a Local Area Network. It is a connection of computers and devices which are confined to a small geographical area. They are therefore relatively small and quite often the network infrastructure will be the property of the organisation.

Top fact: Each device on a network is known as a **node** (e.g. computer, printer, etc.)!

Advantages of Local Area Networks

They enable:

- Digital communication between people
- The sharing of digital information
- The sharing of peripheral devices such as printers and scanners
- Computers to be updated with the latest software from a central point
- Distributed processing – the ability for a single program to be run simultaneously at various computers.

Disadvantages of Local Area Networks

- They require a bit of 'expertise' to install and maintain a large network which can be costly.
- There are a number of security issues from unauthorised access to data.

Measures to secure a network include:

- Passwords – strong passwords use a range of character types
- Not allowing users to install software
- With wireless access, use encryption
- Changing passwords frequently

What is needed to make a LAN?

At least two computers (nodes), each having a **NIC (Network Interface Card** (either wired or wireless)).



The NICs convert the data signals from the nodes into data signals that can be transferred across the network.

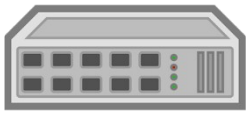
Data Transfer Media:

- Wires
- Wireless Technology



A Hub or a Switch

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A hub is a device that connects nodes together. It makes all connected nodes act as a single segment. It is not considered intelligent as data packets are transmitted across the whole of the network. They are however cheap devices. For example USB hubs are often used in home network installations.



A switch is also a device that connects nodes together. It is considered as an intelligent device as it can create and connect to different segments of the network which reduces network traffic because data packets are sent to the nodes or segment where they are needed rather than being broadcast to the whole network.

A Router

Although not need to form a LAN, a router is a required if the LAN is to connect to the Internet. Routers manage the data packets enabling them to be sent between networks (e.g. over the internet).

WAP – Wireless Access Point

If the LAN is to make use of wireless technology if will require a 'Wireless Access Point'. Wireless technology has many advantages over wired networks:



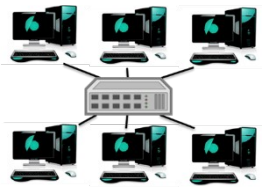
- There is no need to do building work
- You can add new node with ease
- The general public can access network / internet if allowed.

But there are drawbacks. There can be performance issues as a result of signal interference. There can also be security issues if the data being transferred is not encrypted and is intercepted.

Types of Network in a Local Area Network

There are two types of LAN:

- Peer-to-Peer
- Client Server



In a **peer-to-peer** network, all computers have the same status. Data transfer may be very slow because of data collisions and shared processor power. They are only really suitable for small and low traffic networks and as a result are a very popular design for a home network

In a **client-server** network, at least one computer is designated as a server which is usually required to be a high spec machine due to the jobs it will perform (serving the clients).

A server will offer services such as software and data to client PCs (the client will request a service and the server will serve them). The server will also manage the traffic on the network, records what client users do and provide network security (clients will have to sign in and so their identification is known enabling certain rights to be given to certain people). Large networks may have many servers to handle the increased network demand.

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Network Topologies

There are several different ways that LANs can be set up and these 'setups' are known as topologies. The topology that an organisation will choose will be dependent on the money available, the performance required and the expertise at hand.

Bus Topology

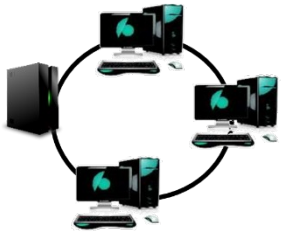
In a bus topology there is only one cable which all nodes connect to.



Advantages	Disadvantages
• Cheap to set up due to little cabling requirements.	• Because data travels both ways along one cable, data collides which slows the network. • If the cable has a fault the whole network may suffer. • Only really useful over small areas.

Ring Topology

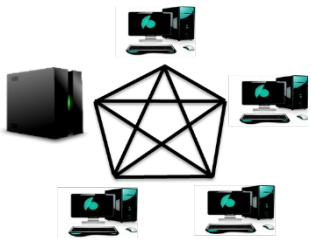
In a ring topology, there is effectively a single cable connecting all nodes, however data travels in one direction, which will mean few data collisions.



Advantages	Disadvantages
• Cheap to set up due to little cabling requirements. • Data travels in only one direction which means fewer data collisions.	• If the cable has a fault the whole network may suffer.

Mesh Topology

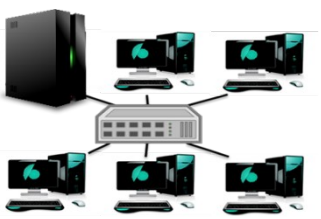
A mesh topology is a fully connected network as each device is connected to all others, which is good because if a cable has a fault, there will be another route available for data to travel.



Advantages	Disadvantages
• If a cable breaks or if there is heavy traffic in one area, there are other routes for the data to take.	• Can be a very expensive set up due to the cabling required.

Star Topology

In a star topology, a hub or a switch will be used to connect all devices, which means that the network performance will be faster



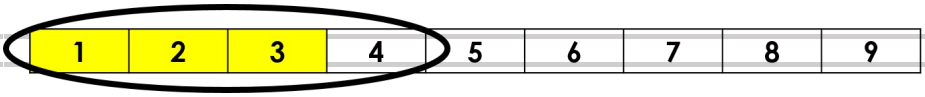
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due to the increase in cabling – there will be more available routes resulting in fewer data collisions.

Advantages	Disadvantages
• All cables connect to a hub or a switch and therefore data doesn't share one cable which results in fewer data collisions and therefore a faster network.	• It is the most expensive topology as it requires a lot of cable and addition network devices (hub/switch).

Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3



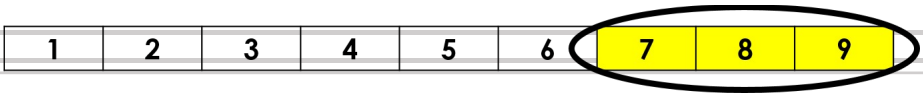
1. Write a definition of a LAN. [3]
2. Describe the differences between a 'Bus Topology' and a 'Ring Topology'. [3]
3. State the devices needed to set up a LAN and describe their roles. [8]

Question Zone 4-6



1. State the devices needed to set up a LAN and describe their roles. [8]
2. Explain the similarities and differences between a HUB and a SWITCH? [3]
3. Explain the differences between a 'Peer-to-Peer' network and a 'Client-Server' network. [6]

Question Zone 7-9



1. Explain the differences between a 'Peer-to-Peer' network and a 'Client-Server' network. [6]
2. Discuss the advantages and disadvantages of networking computers. [6]
3. Discuss the relative advantages and disadvantages of the various network topologies and explain which topology is best suited to a school. [10]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

LAN: *Local Area Network – One which covers a small geographical area and whose equipment is owned by the organisation/individual.*
Node: The name given to any device attached to a network.
Switch*: A device, which connects the computers in a local area network together that is considered intelligent, as it will route data to the correct destination.*

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Stick answer sheet here

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

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